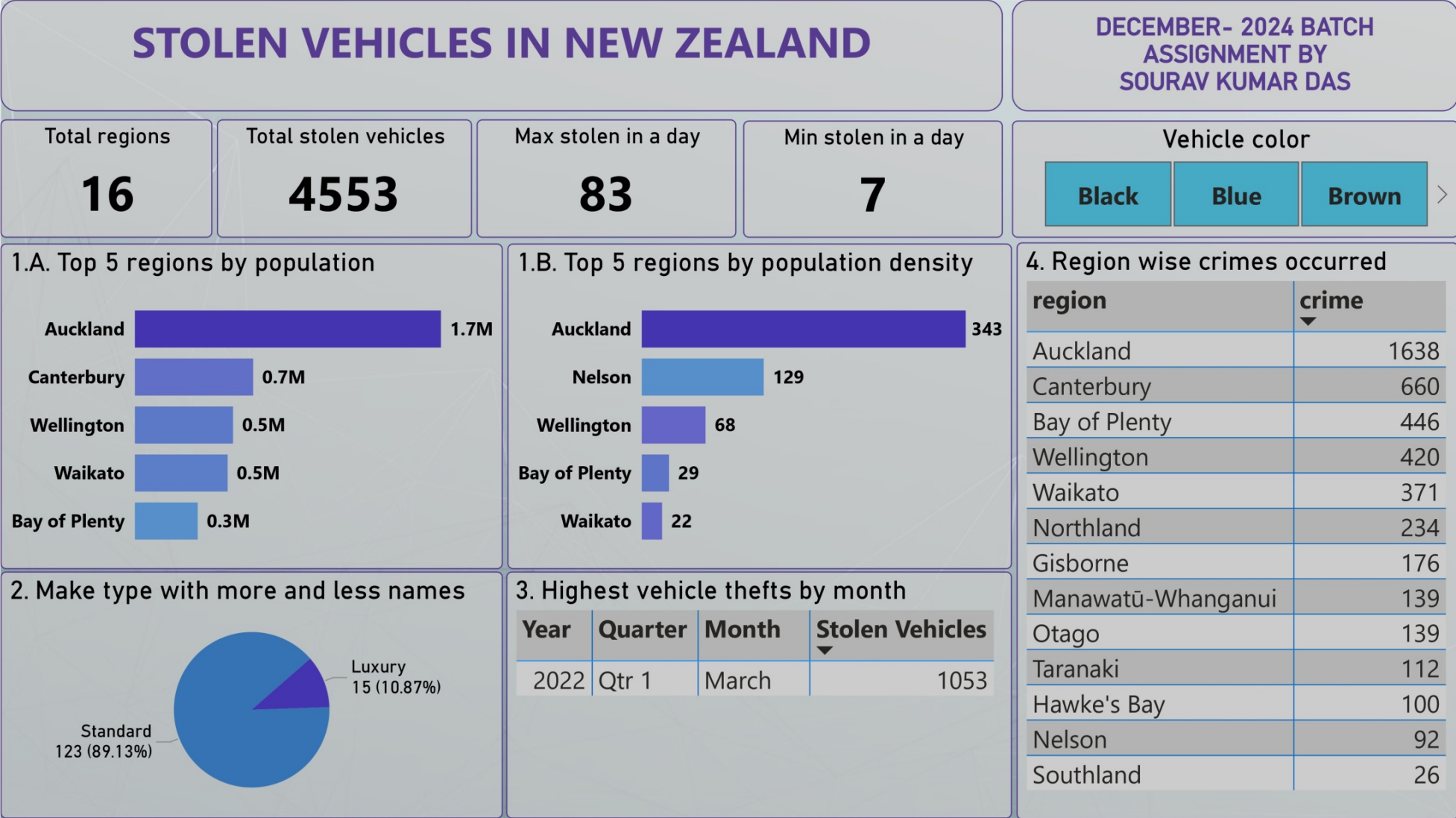


ASSIGNMENT- STOLEN VEHICLES IN NEW ZEALAND

DASHBOARD



## VOLUME INFORMATION

This report is created based on **4553 records**, which covers **16 regions** in New Zealand.

## ASSUMPTION

Total dashboard is made on real given data, so no assumption is made.

## ANALYSIS

### 1. A) Top 5 regions by population:

On dashboard it represents the five most populated regions in New Zealand with their exact population figures:

- **Auckland** has the highest population **1,695,200 (1.7 million)**, making it a prime area for vehicle theft.
- **Canterbury** follows with **655,000 (0.7 million)**, showing a significant drop compared to Auckland.
- **Wellington**- follows with **543,500 (0.5 million)**, showing a moderate population size.
- **Waikato**- follows with **513,800 (0.5 million)**.
- **Bay of Plenty** has the fifth highest population at **347,700 (0.3 million)**.

### 1. B) Top 5 regions by population density:

On Dashboard it represents population density which measures the number of people per square kilometer, providing insights into crowded areas where vehicle theft may be more frequent:

- **Auckland** has the highest population density (**343 people/km<sup>2</sup>**), making it more susceptible to vehicle theft.
- **Nelson** follows with (**129 people/km<sup>2</sup>**), despite its lower overall population.
- **Wellington** has a density of (**68 people/km<sup>2</sup>**), suggesting that even mid-sized cities face security concerns.
- **Bay of Plenty** (**29 people/km<sup>2</sup>**) and **Waikato** (**22 people/km<sup>2</sup>**) have lower densities, possibly correlating to a lower theft risk compared to more crowded regions.

## 2) Make type with more and less names:

- The Standard make type has the most make names, with **123 (89.13%)** makes.
- The Luxury make type has the fewest make names, with **15 (10.87%)** makes.

## 3) Highest vehicle thefts by month:

The maximum crime happened in **March 2022**, with **1053** stolen vehicles. This falls under **Quarter 1 (Q1)** of the year **2022**.

## 4) Region wise crimes occurred:

Vehicle thefts have been recorded in **13 out of 16 regions** in New Zealand. Below is a detailed breakdown:

1. **Auckland (1,638 thefts)** – The region with the highest number of vehicle thefts.
2. **Canterbury (660 thefts)** – The second-highest vehicle theft cases recorded.
3. **Bay of Plenty (446 thefts)** – The third-highest region for stolen vehicles.
4. **Wellington (420 thefts)** – Reports a significant number of vehicle thefts.
5. **Waikato (371 thefts)** – A moderately affected region by vehicle thefts.
6. **Northland (234 thefts)** – Reports a considerable number of stolen vehicles.
7. **Gisborne (176 thefts)** – Experiences a noticeable amount of vehicle thefts.
8. **Manawatū-Whanganui (139 thefts)** – A region with a moderate theft rate.
9. **Otago (139 thefts)** – Reports the same theft count as Manawatū-Whanganui.
10. **Taranaki (112 thefts)** – Has over a hundred recorded stolen vehicles.
11. **Hawke's Bay (100 thefts)** – Records exactly 100 vehicle theft cases.
12. **Nelson (92 thefts)** – A lower theft count compared to other regions.
13. **Southland (26 thefts)** – The region with the lowest reported vehicle thefts.

## 5) Key Insights at a Glance:

- **Total Stolen Vehicles** - Displays the total number of stolen vehicles recorded during the year 2021-2022.
- **Total Regions** – Shows the total number of regions in New Zealand where the crimes occurred.
- **Maximum Stolen in a Day** – Represents the highest number of vehicle thefts in a single day.
- **Minimum Stolen in a Day** – Indicates the lowest number of vehicle thefts in a single day.

The dashboard also contains a slicer named as '**Stolen Vehicle Color**', which utilizes the **color column** from the dataset. This column contains the color of stolen vehicles and is used to dynamically filter two visual cards: "**Maximum Stolen in a Day**" and "**Minimum Stolen in a Day**." When a specific color is selected from the slicer, these cards update accordingly to display the highest and lowest number of stolen vehicles for that chosen color, allowing for a more detailed and interactive analysis of theft patterns.

## SUGGESTION

### 1. Enhancing Security in High-Theft Regions

- Auckland, Canterbury, and Bay of Plenty, Wellington and Waikato should implement stronger vehicle security measures.
- More surveillance, police patrols, and awareness campaigns could help reduce vehicle theft.

### 2. Targeted Law Enforcement Strategies

Regions with high theft density (Auckland, Nelson, Wellington) should have **specific law enforcement strategies**, such as:

- More CCTV coverage in high-risk areas.
- Public awareness about parking in secure areas.
- Vehicle tracking system adoption.

### 3. Vehicle Safety Measures

- Car owners should consider installing **GPS trackers, alarms, and steering locks**.
- The government can introduce **mandatory anti-theft measures** for high-risk areas.

#### 4. Investigating Crime Patterns

- Analyzing the preference for **specific vehicle colors** could help law enforcement **track stolen vehicles more effectively**.

#### 5. Public Awareness Campaigns

- Raising awareness about car theft hotspots can **help car owners take precautionary measures**.
- Encouraging the **use of secured parking** and **vehicle immobilizers**.

### SUMMARY

The data highlights a significant concern regarding vehicle theft in New Zealand, particularly in **Auckland and other highly populated regions**. While standard vehicles remain the most targeted, law enforcement and preventive measures can help curb these crimes. Strengthening vehicle security and public awareness will be key to reducing vehicle theft incidents.

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